UNIT

II

ENGINEERING PHYSICS • DIGITAL NOTES

Crystallography and X-ray Diffraction

Space lattice • Basis • Unit cell • Bravais lattices • Packing fraction • Miller indices • Bragg's law • Diffractometer • Laue and powder methods

QUICK SUMMARY • What you will learn

- How atoms are arranged in crystals and how to describe that arrangement in a language of lattices and cells
- How to label planes with Miller indices and calculate the distance between them
- How X-rays reveal the inside of a crystal (Bragg's law, Laue and powder methods)



Fig. 1: Roadmap of Unit II

1 Space Lattice, Basis and Unit Cell

Crystalline solids	Amorphous solids
Long-range, periodic order of atoms	No long-range order (short-range only)
Sharp melting point; anisotropic	Softens over a range; isotropic
Examples: NaCl, Cu, diamond, quartz	Examples: glass, rubber, plastics

Space lattice: an infinite array of points in space such that every point has **identical surroundings**.

Basis: the atom or group of atoms attached to every lattice point.

Crystal structure = Lattice + Basis.

Unit cell: the smallest volume (parallelepiped) which, repeated in three dimensions by translation, builds the whole crystal.

Primitive cell: a unit cell with lattice points only at its corners (one lattice point per cell). Non-primitive cells have extra points at the body or face centres.

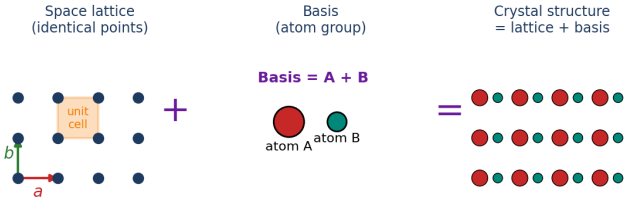


Fig. 2: Lattice, basis and crystal structure (2-D view)

- Edges a, b, c and angles α (between b, c), β (between c, a) and γ (between a, b) are the six **lattice parameters**.
- Translation vector: $\mathbf{T = n_1 a + n_2 b + n_3 c}$ (n_1, n_2, n_3 integers) takes any lattice point to an equivalent one.

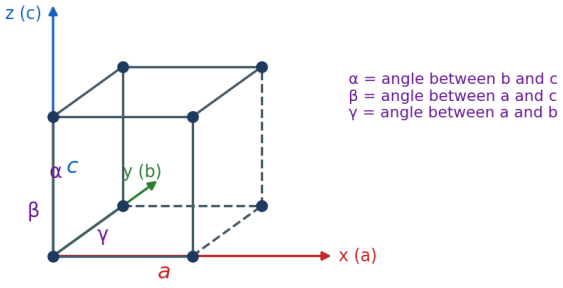


Fig. 3: Unit cell and lattice parameters

2 Crystal Systems and Bravais Lattices

On the basis of the relations among a, b, c and α, β, γ, all crystals fall into **7 crystal systems**. Adding centring (P = primitive, I = body centred, F = face centred, C = base centred) gives exactly **14 Bravais lattices** (Bravais, 1848).

System	Axes	Angles	Bravais lattices	Examples
Cubic	a = b = c	α = β = γ = 90°	P, I, F (3)	Cu, NaCl, diamond
Tetragonal	a = b ≠ c	α = β = γ = 90°	P, I (2)	White tin, TiO_2
Orthorhombic	a ≠ b ≠ c	α = β = γ = 90°	P, C, I, F (4)	$BaSO_4$, rhombic sulphur
Monoclinic	a ≠ b ≠ c	α = γ = 90° ≠ β	P, C (2)	Gypsum, monoclinic S
Triclinic	a ≠ b ≠ c	α ≠ β ≠ γ ≠ 90°	P (1)	$CuSO_4 \cdot 5H_2O$
Trigonal (rhombohedral)	a = b = c	α = β = γ ≠ 90°	P (1)	Calcite, Bi
Hexagonal	a = b ≠ c	α = β = 90°, γ = 120°	P (1)	Zn, Mg, graphite

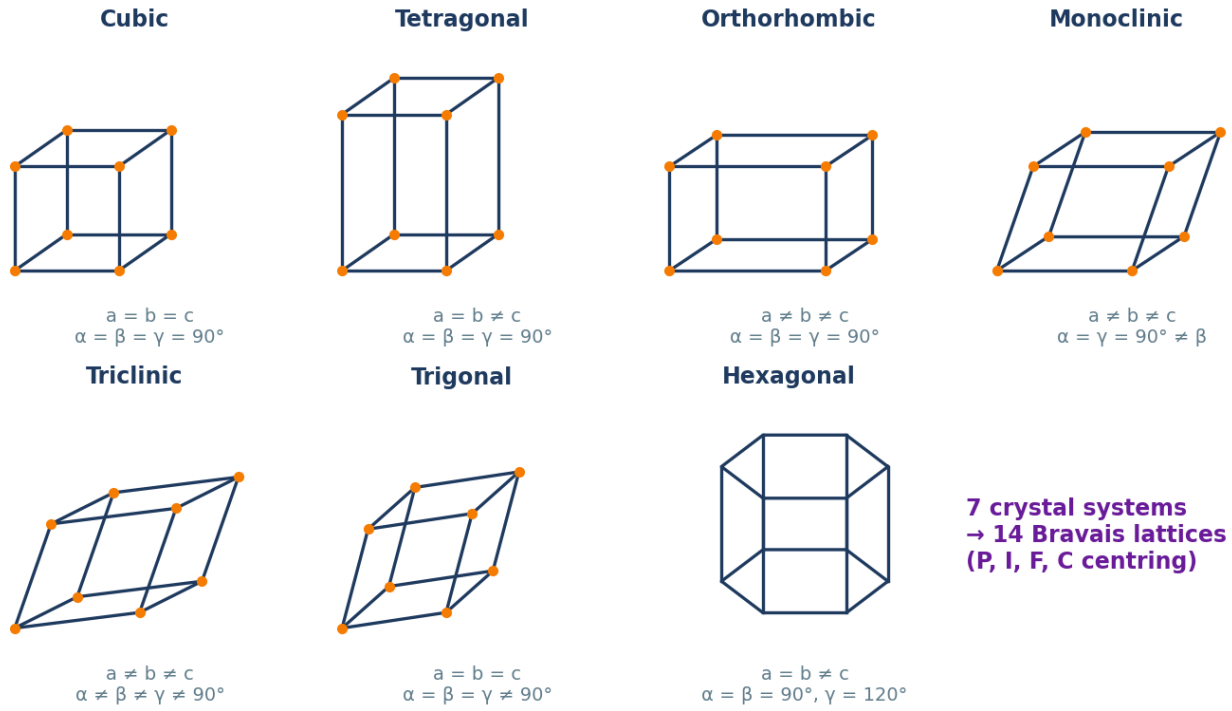


Fig. 4: Unit cells of the seven crystal systems

REMEMBER • Counting trick

3 + 2 + 4 + 2 + 1 + 1 + 1 = **14**. Only the cubic system has all three of P, I and F; only orthorhombic has all four types.

3 Cubic Structures: Coordination Number and Packing Fraction

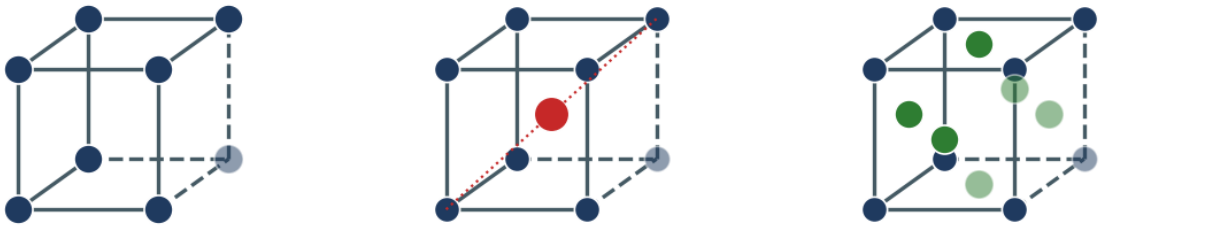


Fig. 5: The three cubic Bravais lattices and the number of atoms they contain

DEFINITION • Coordination number (N)

The number of nearest neighbours of an atom in the crystal.

DEFINITION • Atomic packing fraction (APF)

APF = (volume of atoms in a unit cell) / (volume of the unit cell) = $n \cdot (4/3)\pi r^3 / a^3$, where n = atoms per cell.

Property	SC	BCC	FCC
Atoms per cell (n)	1	2	4
Coordination number	6	8	12
Touching atoms along	Cube edge	Body diagonal	Face diagonal
Relation of r and a	a = 2r	a = 4r/√3	a = 4r/√2 = 2√2 r
Nearest-neighbour distance	a	√3 a/2	a/√2
Packing fraction	π/6 = 0.52 (52%)	√3π/8 = 0.68 (68%)	√2π/6 = 0.74 (74%)
Examples	Polonium	Fe, Na, Cr	Cu, Al, Ag, Au

SC (r = a/2, n = 1)

$$APF = \frac{1 \times \frac{4}{3}\pi\left(\frac{a}{2}\right)^3}{a^3} = \frac{\pi}{6} = 0.52$$

BCC (r = √3a/4, n = 2)

$$APF = \frac{2 \times \frac{4}{3}\pi\left(\frac{\sqrt{3}a}{4}\right)^3}{a^3} = \frac{\sqrt{3}\pi}{8} = 0.68$$

FCC (r = a/2√2, n = 4)

$$APF = \frac{4 \times \frac{4}{3}\pi\left(\frac{a}{2\sqrt{2}}\right)^3}{a^3} = \frac{\pi}{3\sqrt{2}} = 0.74$$

EXAMPLE • Copper

Cu is FCC with r = 1.278 Å. a = 2√2 r = **3.615 Å**. Density $\rho = nM/(N_A a^3) = 4 \times 63.5\times10^{-3} / (6.022\times10^{23} \times (3.615\times10^{-10})^3) \approx$ **8.9 × 10³ kg/m³**.

Beyond the simple cubic family

Structure	Atoms / cell	Coordination no.	Packing fraction	Examples
HCP (c/a = 1.633)	6 (2 per primitive cell)	12	0.74	Zn, Mg, Ti
Diamond cubic	8	4	0.34	C (diamond), Si, Ge
NaCl (rock salt)	4 Na^+ + 4 Cl^-	6	—	NaCl, KCl, MgO

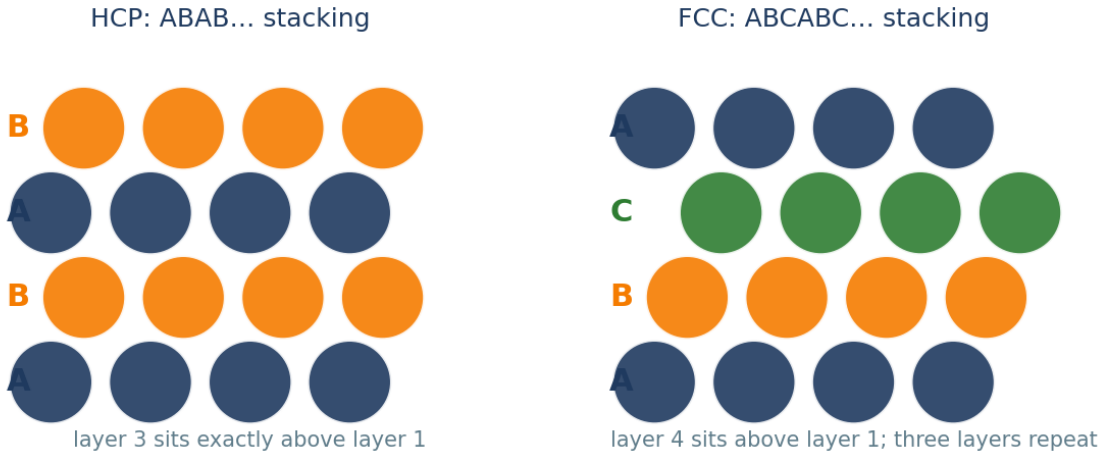


Fig. 6: Close-packed layers: HCP repeats every two layers, FCC every three

KEY IDEA • Remember

FCC (and HCP) are **close-packed** structures: 74% of the space is filled and 26% is empty. Tight packing is why many metals (Cu, Al, Au) are ductile.

4 Miller Indices

DEFINITION • Miller indices (h k l)

A set of three smallest integers, proportional to the reciprocals of the intercepts a plane makes on the crystallographic axes, used to label planes in a crystal.



Fig. 7: Procedure for finding Miller indices

EXAMPLE • Worked example

A plane cuts intercepts 2a, 3b, 6c. Intercepts (2, 3, 6) → reciprocals (1/2, 1/3, 1/6) → multiply by LCM 6 → **(3 2 1)**.
A plane parallel to the z-axis has intercept ∞ on z, so l = 1/∞ = 0, e.g. intercepts (1, 1, ∞) → **(1 1 0)**.

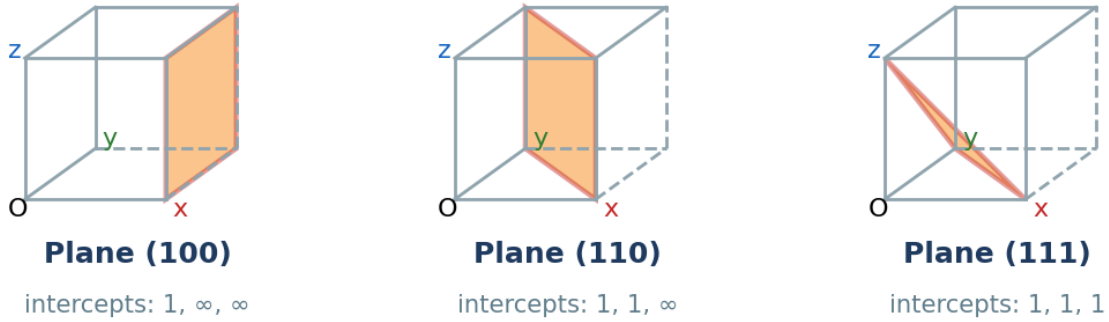


Fig. 8: The (100), (110) and (111) planes of a cubic cell

- **Parallel planes** have the same Miller indices.
- A negative intercept is written with a bar on top, e.g. ($1\bar{1}$ 0) is read as “one-bar one zero”.
- (h k l) = one plane; {h k l} = family of equivalent planes; [u v w] = a direction; ⟨u v w⟩ = family of directions.
- Larger indices mean closer spaced planes and lower atomic density per plane.

Separation between successive (hkl) planes

Let the plane nearest to the origin make intercepts a/h, b/k, c/l (points A, B, C) and let ON = d be the perpendicular from the origin. If the normal makes angles α, β, γ with the axes, then cosα = d/(a/h) = hd/a, cosβ = kd/b, cosγ = ld/c. For orthogonal axes $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$, which gives the results below.

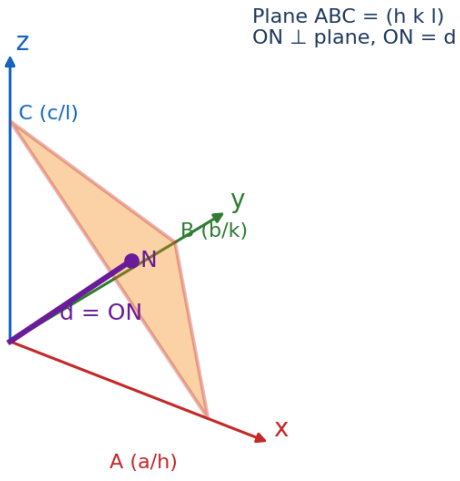


Fig. 9: Geometry for d(hkl)

General (orthogonal axes)	$\frac{1}{d_{hkl}^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$
Cubic crystal (a = b = c)	$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$
Ratio of spacings in cubic	$d_{100} : d_{110} : d_{111} = 1 : \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{3}}$

EXAMPLE • Numerical

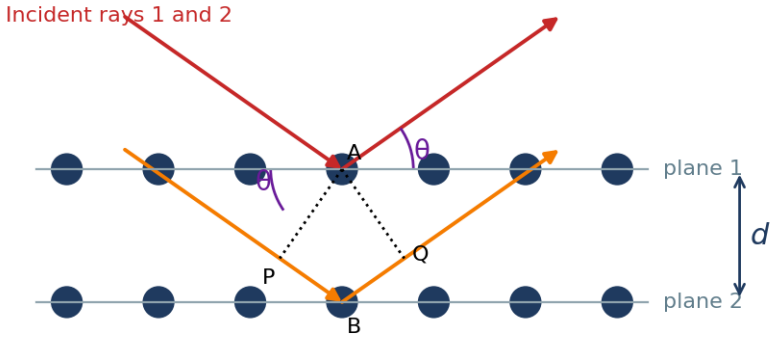
For Cu (a = 3.615 Å): d_{111} = 3.615/√3 = **2.087 Å**; d_{200} = 3.615/2 = **1.808 Å**; d_{220} = 3.615/√8 = **1.278 Å**.

5 X-ray Diffraction and Bragg's Law

DEFINITION • X-rays

Electromagnetic waves of wavelength 0.1–10 Å, produced when fast electrons strike a metal target. The spectrum has a continuous part (shortest wavelength λ_{min} = 12400/V Å, with V in volts) and sharp characteristic lines (Kα, Kβ) of the target element.

X-rays have wavelengths of about **0.1–10 Å**, comparable to interatomic spacings in crystals, so a crystal behaves like a natural three-dimensional diffraction grating for X-rays. Bragg (1913) treated diffraction as "reflection" from successive atomic planes.



Path difference = PB + BQ = 2d sinθ → 2d sinθ = nλ

Fig. 10: Reflection of X-rays from two successive planes of spacing d

Rays 1 and 2 reflect from planes a distance d apart at glancing angle θ. The extra path travelled by ray 2 is PB + BQ = d sinθ + d sinθ. For constructive interference this must be a whole number of wavelengths.

Bragg's law	$2d\sin\theta = n\lambda \qquad n = 1, 2, 3, \ldots$
Cubic crystal (n = 1)	$\sin^2\theta = \frac{\lambda^2}{4a^2}(h^2 + k^2 + l^2)$
Maximum order possible	$n_{max} = \frac{2d}{\lambda} \quad (\sin\theta \le 1 \Rightarrow \lambda \le 2d)$

REMEMBER • Points to note

- Bragg's law works only if λ ≤ 2d: visible light cannot be diffracted by atomic planes.
- θ is the **glancing angle** (angle with the plane), not with the normal; the diffracted beam is deviated by 2θ from the incident beam.
- Unlike ordinary reflection, Bragg reflection occurs only at specific angles.

EXAMPLE • Numerical

NaCl planes have d = 2.82 Å. For X-rays of λ = 1.54 Å: first order sinθ = 1.54/(2 × 2.82) = 0.273, so θ = **15.8°**. Second order: sinθ = 0.546, θ = **33.1°**.

6 X-ray Diffractometer

A diffractometer records the intensity of diffracted X-rays electronically against the angle 2θ, usually for a powder sample. It replaces the photographic film with a counter, so data are fast, accurate and digital.

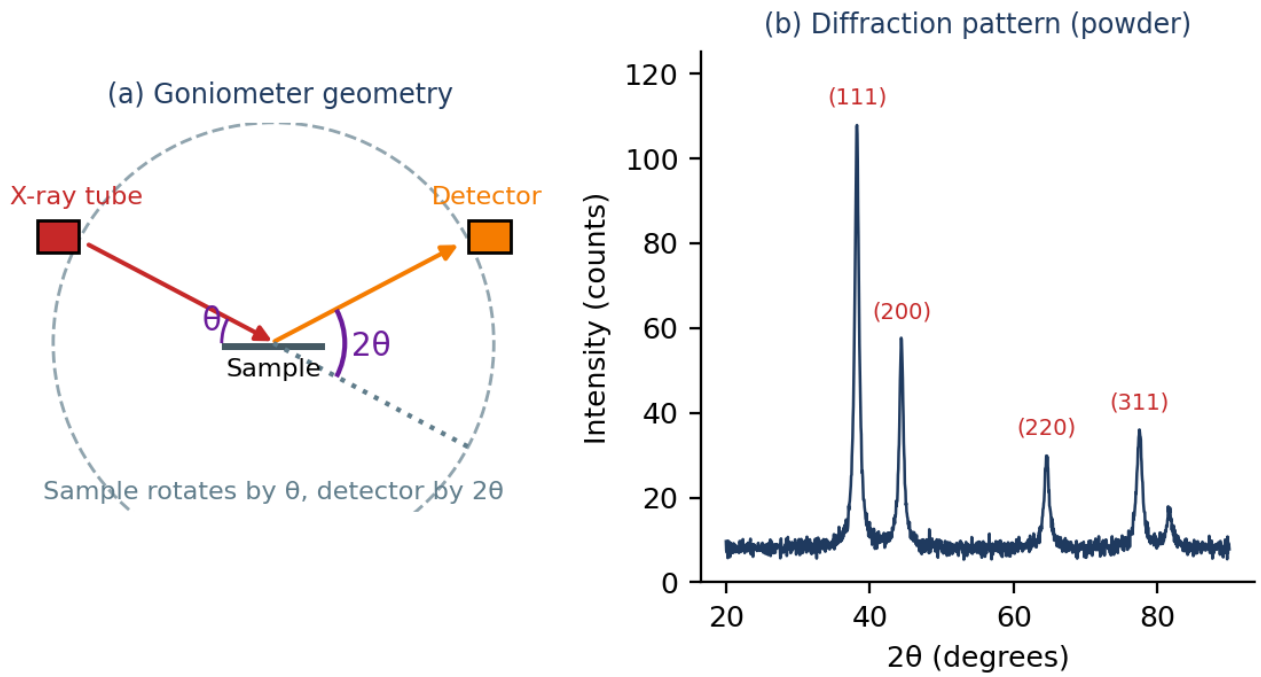


Fig. 11: Goniometer geometry and a typical powder pattern (peak positions give d, peak heights give atomic arrangement)

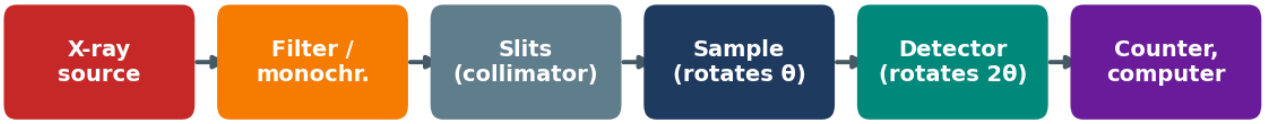


Fig. 12: Block diagram of an X-ray diffractometer

Part	Function
X-ray tube	Fast electrons strike a metal target (Cu, λ(Kα) = 1.54 Å; Mo) to produce characteristic X-rays
Filter / monochromator	Ni foil removes Kβ radiation so that the beam is almost monochromatic
Slits (collimator)	Make a narrow, nearly parallel beam; also limit scattering
Sample stage (goniometer)	Holds the powder flat; rotates through θ at constant angular speed
Detector	Scintillation or proportional counter rotating at 2θ collects the diffracted beam
Ratemeter + computer	Convert pulses to counts; plot intensity versus 2θ

- **Output:** peaks at angles that satisfy Bragg's law; compare with standard database (JCPDS/ICDD) to identify phases.
- **Uses:** phase identification, lattice parameters, crystallite size (Scherrer: $D = K\lambda / \beta\cos\theta$), strain, percentage crystallinity.

7 Crystal Structure Determination: Laue and Powder Methods

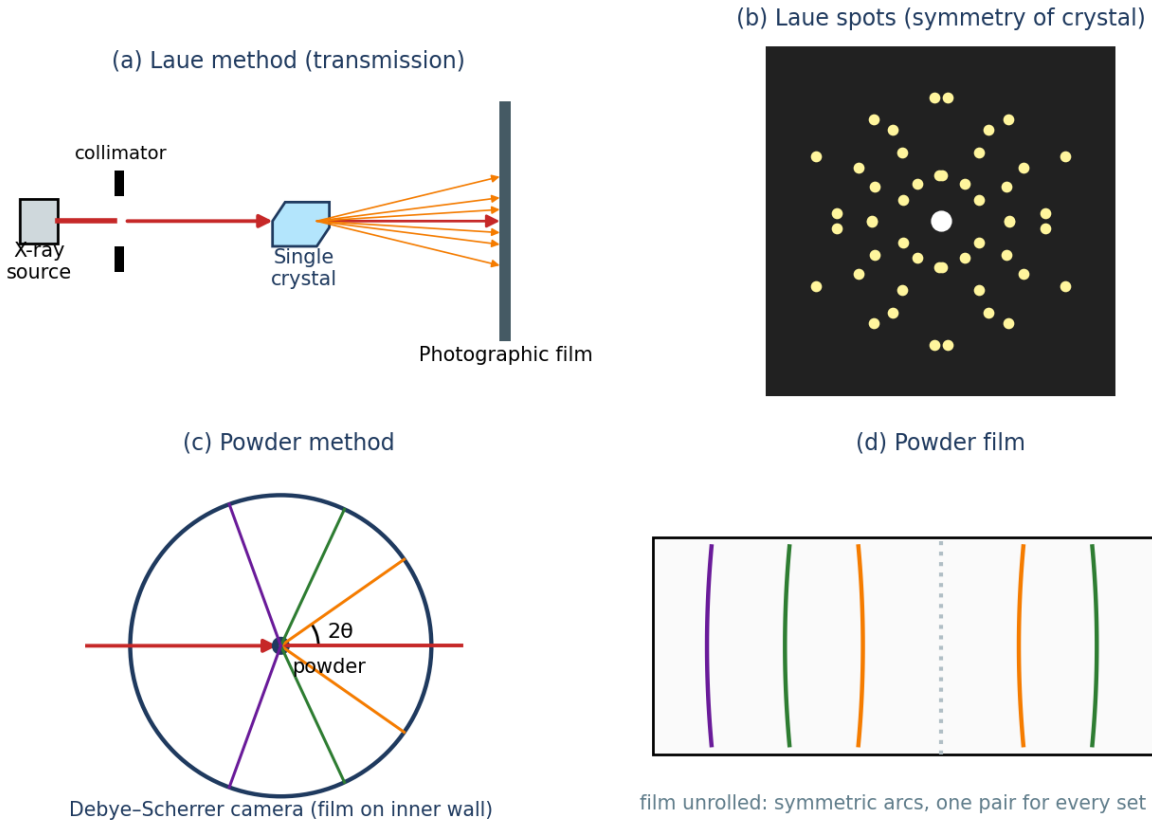


Fig. 13: Laue method (a, b) and Debye–Scherrer powder method (c, d)

Feature	Laue method	Powder (Debye-Scherrer) method
Sample	Single crystal, fixed	Fine powder or polycrystal (random orientation)
X-rays	**White** (continuous λ)	**Monochromatic**
Idea	For each plane set, the crystal picks the λ that satisfies 2d sinθ = nλ	Among many crystallites, some are always oriented at the Bragg angle for each (hkl)
Pattern	Spots (symmetric dot pattern)	Concentric cones → arcs on film
Gives	Crystal symmetry and orientation; checks perfection	Lattice type, d-spacings, lattice constant
Variants	Transmission and back-reflection	Film camera or diffractometer

Powder method: step-by-step

- Measure the distance S between a symmetric pair of arcs on the film. For camera radius R: $4\theta = S/R$ (in radians), so $\theta = S/4R$.
- Find d from Bragg's law, $d = \lambda/(2\sin\theta)$, for each line.
- Compute $\sin^2\theta$ for each line and compare the ratios with the table to find the lattice type.
- Index the lines (h k l) and compute the lattice constant $a = \lambda\sqrt{(h^2+k^2+l^2)} / (2\sin\theta)$.

Cubic lattice	Allowed reflections	$\sin^2\theta$ ratio ($\propto h^2+k^2+l^2$)
Simple cubic	all (hkl)	1 : 2 : 3 : 4 : 5 : 6 : 8 : 9 ... (7 missing)
BCC	h + k + l even	2 : 4 : 6 : 8 : 10 : 12 ... = 1 : 2 : 3 : 4 : 5 : 6 ...
FCC	h, k, l all odd or all even	3 : 4 : 8 : 11 : 12 : 16 ...

KEY IDEA • Quick identification
Take the ratio of the first two lines: **0.75 (3:4)** suggests FCC; **1:2 pattern** with the first line = 2 suggests BCC; the series 1:2:3:4:5:6 with a missing 7 suggests SC.

APPLICATIONS • Applications of X-ray diffraction

- Identifying unknown materials and phases by their “fingerprint” pattern.
- Finding lattice constants, crystal size, strain and defects in metals and nanomaterials.
- Determining structure of proteins and DNA (X-ray crystallography); quality control in pharmaceuticals and semiconductors.

8 Worked Problems

EXAMPLE • 1. Packing

BCC iron has a = 2.87 Å. r = √3a/4 = **1.24 Å**; nearest-neighbour distance = 2r = 2.48 Å; density = 2 × $55.85\times10^{-3}/(6.022\times10^{23} \times (2.87\times10^{-10})^3)$ = **7.85×10^3 kg/m³**.

EXAMPLE • 2. Miller indices

A plane has intercepts a, b/2, c/3 → reciprocals 1, 2, 3 → **(1 2 3)**. Its spacing in a cubic crystal with a = 4.2 Å: d = 4.2/√14 = **1.12 Å**.

EXAMPLE • 3. Bragg’s law

For d = 1.1 Å, the maximum order for λ = 0.5 Å: n_{max} = 2d/λ = 4.4 → **4**. First-order glancing angle: sinθ = 0.5/2.2 = 0.227, θ = **13.1°**.

EXAMPLE • 4. Powder pattern

A cubic powder (λ = 1.54 Å) shows sin²θ = 0.1365, 0.1820, 0.3640, 0.5005. Dividing by 0.0455 gives **3 : 4 : 8 : 11** → **FCC** with lines (111), (200), (220), (311). a = λ√3/(2 sinθ) = 1.54 × 1.732 / (2 × 0.3695) = **3.61 Å** (copper).

EXAM CORNER • Try yourself (answers in brackets)

- Aluminium is FCC with a = 4.05 Å. Find the atomic radius and density. [1.43 Å; 2.7 × 10³ kg/m³]
- Find the Miller indices of a plane with intercepts 3a, 2b, c. [(2 3 6)]
- Silicon has a = 5.43 Å. Find d_{220}. [1.92 Å]
- X-rays of λ = 1.54 Å are reflected from planes with d = 2.5 Å. Find the first-order Bragg angle. [17.9°]
- The first two powder lines of a cubic crystal have sin²θ in the ratio 1 : 2. Which lattice is this likely to be, and what are the indices of the lines? [BCC; (110), (200)]

Key terms at a glance

Term	Meaning
Lattice point	A point with identical surroundings to every other point of the lattice
Primitive vectors	Smallest non-coplanar translation vectors a, b, c of the lattice
Glancing angle	Angle θ between the incident X-ray beam and the atomic plane
Systematic absence	A reflection missing because of centring (e.g. BCC needs h+k+l even)
Anisotropy	Dependence of physical properties on crystal direction

Diffraction by X-rays, electrons and neutrons

	X-rays	Electrons	Neutrons
Interacts with	Electron cloud	Electrostatic potential (strong)	Nuclei (and magnetic moments)
Penetration	Moderate (μm–mm)	Very small (surface, thin films)	Large (bulk)
Wavelength from	Target metal / tube voltage	λ = h/√(2meV)	λ = h/√(2m_nE)
Best for	General structure	Surfaces, thin films, TEM	Light atoms (H), magnetic structure

9 Formula Sheet and Exam Corner

Topic	Key formulae
Lattice	Crystal = lattice + basis; 7 systems, 14 Bravais lattices
Atoms / cell, N	SC: 1, 6; BCC: 2, 8; FCC: 4, 12
APF	SC 0.52, BCC 0.68, FCC 0.74. Density $\rho = nM / (N_A a^3)$
Miller indices	Reciprocals of intercepts → smallest integers (h k l)
Plane spacing	$d = a/\sqrt{(h^2+k^2+l^2)}$ (cubic); $1/d^2 = h^2/a^2 + k^2/b^2 + l^2/c^2$
Bragg	$2d \sin\theta = n\lambda$; $a = \lambda\sqrt{(h^2+k^2+l^2)}/(2 \sin\theta)$
Powder film	$\theta = S / 4R$ (radians)

EXAM CORNER • Short questions (2–3 marks)

- Define space lattice, basis and unit cell.
- List the seven crystal systems with their axial relations.
- Define coordination number and packing fraction.
- Calculate the packing fraction of an FCC structure.
- What are Miller indices? Find the Miller indices of a plane with intercepts 2a, 3b, ∞.
- State Bragg’s law. Why can visible light not be used?

EXAM CORNER • Long questions (10 marks)

- Describe the 14 Bravais lattices and 7 crystal systems.
- Derive the packing fractions of SC, BCC and FCC.
- Derive an expression for the interplanar spacing in terms of Miller indices for a cubic lattice.
- Derive Bragg’s law. Explain the working of an X-ray diffractometer with a neat sketch.
- Explain Laue’s method and the powder method for crystal structure determination.

EXAM CORNER • Quick MCQs

- 1. Number of atoms in an FCC unit cell: (a) 2 (b) 4 (c) 6.
- 2. Which has the highest packing fraction? (a) SC (b) BCC (c) FCC.
- 3. d_{110} in a cubic crystal of edge a = (a) a (b) $a/\sqrt{2}$ (c) $a/\sqrt{3}$.
- 4. The Laue method uses (a) monochromatic X-rays (b) white X-rays.
- **Answers:** 1-b, 2-c, 3-b, 4-b.

QUICK SUMMARY

- A crystal is a lattice with a basis; 14 Bravais lattices belong to 7 systems.
- Packing: SC 52%, BCC 68%, FCC 74%.
- $d_{hkl} = a/\sqrt{(h^2+k^2+l^2)}$; Bragg: $2d \sin\theta = n\lambda$. Laue uses white X-rays on a single crystal, the powder method uses monochromatic X-rays on powder.